



Java TreeSet

[< Previous](#)[Next >](#)

Java TreeSet

A **TreeSet** is a collection that stores unique elements **in sorted order**.

It is part of the `java.util` package and implements the [Set](#) interface.

Tip: Unlike **HashSet**, which has no order, **TreeSet** keeps its elements sorted automatically.

Create a TreeSet

Example

Create a **TreeSet** object called **cars** that will store strings:

```
import java.util.TreeSet; // Import the TreeSet class

TreeSet<String> cars = new TreeSet<>();
```



Add Elements

To add elements to a **TreeSet**, use the **add()** method:

Example

```
import java.util.TreeSet;

public class Main {
    public static void main(String[] args) {
        TreeSet<String> cars = new TreeSet<>();
        cars.add("Volvo");
        cars.add("BMW");
        cars.add("Ford");
        cars.add("BMW"); // Duplicate
        cars.add("Mazda");

        System.out.println(cars);
    }
}
```

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Output: The elements will be sorted automatically (e.g., [BMW, Ford, Mazda, Volvo]).

Note: Duplicates like "BMW" will only appear once.

Check if an Element Exists



Example

```
cars.contains("Mazda");
```

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Remove an Element

Use `remove()` to remove an element:

Example

```
cars.remove("Volvo");
```

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Remove All Elements

Use `clear()` to remove all elements:

Example

```
cars.clear();
```



TreeSet Size

Use `size()` to count how many unique elements are in the set:

Example

```
cars.size();
```

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Note: Duplicate values are not counted - only unique elements are included in the size.

Loop Through a TreeSet

Loop through the elements of a `TreeSet` with a **for-each** loop:

Example

```
TreeSet<String> cars = new TreeSet<>();  
// add elements...  
  
for (String i : cars) {  
    System.out.println(i);  
}
```

Try it Yourself »



Example

```
import java.util.TreeSet;

public class Main {
    public static void main(String[] args) {
        TreeSet<Integer> numbers = new TreeSet<>();
        numbers.add(40);
        numbers.add(10);
        numbers.add(30);
        numbers.add(20);

        for (int n : numbers) {
            System.out.println(n);
        }
    }
}
```

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Output: The numbers will be printed in sorted order (10, 20, 30, 40).

HashSet vs TreeSet

Feature	HashSet	TreeSet
Order	No guaranteed order	Sorted (natural order)



Tip: Use [HashSet](#) when you care about speed, and [TreeSet](#) when you need sorted elements.

The var Keyword

From Java 10, you can use the [var](#) keyword to declare a [TreeSet](#) variable without writing the type twice. The compiler figures out the type from the value you assign.

This makes code shorter, **but many developers still use the full type for clarity**. Since [var](#) is valid Java, you may see it in other code, so it's good to know that it exists:

Example

```
// Without var
TreeSet<String> cars = new TreeSet<String>();

// With var
var cars = new TreeSet<String>();
```

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The Set Interface

Note: Sometimes you will see both [Set](#) and [TreeSet](#) in Java code, like this:



```
Set<String> cars = new TreeSet<>();
```

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This means the variable (cars) is declared as a Set (the interface), but it stores a **TreeSet** object (the actual set). Since **TreeSet** implements the **Set** interface, this is possible.

It works the same way, but some developers prefer this style because it gives them more flexibility to change the type later.

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